AQA

A-Level
Economics

7136/2 Paper 2 National And International Economy
Predicted Paper Final Mark scheme

7136
Predicted Paper

Version/Stage: v1.0   [Set C — Advanced]

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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This is a predicted paper intended for revision purposes only. It is not an official AQA publication. The mark scheme follows the style and structure of AQA published materials to support realistic practice.

Annotations

Annotation	Description
?	Unclear
AN	Analysis
APP	Application
BOD	Benefit of the doubt
EVAL	Evaluation
Highlight	Highlight
IR	Irrelevant
KU	Knowledge and understanding
NAQ	Not answered question
REP	Repetition
SEEN	Indicates that the point has been noted, but no credit has been given.
Tick	Correct point
Cross	Incorrect point

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 — Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best-fit approach for defining the level and then use the variability of the response to help decide the mark within the level.

Step 2 — Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Levels of response marking grid — 25-mark questions

The following generic grid should be used when marking all 25-mark questions (questions 0 4, 0 8, 1 0, 1 2 and 1 4 on paper 7136/2).

Level of response	Response	Max 25 marks
5	Sound, focused analysis and well-supported evaluation that: • is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response • includes well-focused analysis with clear, logical chains of reasoning • includes supported evaluation throughout the response and in a final conclusion.	21–25 marks
4	Sound, focused analysis and some supported evaluation that: • is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes some good application of relevant economic principles to the given context and, where appropriate, some good use of data to support the response • includes some well-focused analysis with clear, logical chains of reasoning • includes some reasonable, supported evaluation.	16–20 marks
3	Some reasonable analysis but generally unsupported evaluation that: • focuses on issues that are relevant to the question, showing satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response • includes some reasonable analysis but which might not be adequately developed or becomes confused in places • includes fairly superficial evaluation; there is likely to be some attempt to make relevant judgements but these aren't well-supported by arguments and/or data.	11–15 marks
2	A fairly weak response with some understanding that: • includes some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • includes some limited application of relevant economic principles to the given context and/or data to the question • includes some limited analysis but it may lack focus and/or become confused • includes some evaluation which is weak and unsupported.	6–10 marks
1	A very weak response that: • includes little relevant knowledge and understanding of economic terminology, concepts and principles • includes application to the given context which is, at best, very weak • includes attempted analysis which is weak and unsupported.	1–5 marks

Section A

Context 1 **Total for this Context: 40 marks**

Global supply chains and the rise of protectionism

0 1 Using the data in Extract A (Figure 1), calculate the percentage-point change in the US average applied tariff rate between 2015 and 2024. **[2 marks]**

Calculation:

5.1 – 1.6 = **3.5 percentage points** (increase)

Response	**Max 2 marks**
For the correct answer (3.5 percentage points (3.5pp increase)) with correct working shown.	2 marks
For correct working with an arithmetic slip; OR correct numerical answer without working shown.	1 mark

0 2 Explain how the data in Extract A (Figure 2) suggest that the world economy has entered a period of slower globalisation since the 2008 financial crisis. **[4 marks]**

Response	**Max 4 marks**
• includes evidence that clearly shows that the world economy has entered a period of slower globalisation since the 2008 crisis. • clearly explains how this data is evidence that the world economy has entered a period of slower globalisation since the 2008 crisis.	4 marks
• includes evidence that shows that the world economy has entered a period of slower globalisation since the 2008 crisis. • unclear explanation of how this data is evidence that the world economy has entered a period of slower globalisation since the 2008	3 marks
• includes evidence that shows that the world economy has entered a period of slower globalisation since the 2008 crisis. • limited or no explanation of how this data is evidence that the world economy has entered a period of slower globalisation since the	2 marks
• includes evidence that does not clearly show that the world economy has entered a period of slower globalisation since the 2008 cri • weak or no explanation.	1 mark

Relevant issues include:

- definition of globalisation — deepening integration of national economies via trade, investment and capital flows
- world goods trade volume growth averaged 7.2% (2000–2007) but fell to 3.2% (2010–2019) — less than half its pre-crisis pace
- cross-border FDI flows collapsed from 15.4% growth (2000–2007) to near-zero (0.1%) in the 2010s, and fell sharply (–9.5%) in 2023
- although global GDP growth has also slowed, the ratio of trade growth to GDP growth has fallen — indicating less trade per unit of income, i.e. slowing integration
- 2023 particularly stark: trade volume growth of just 0.8% while GDP grew 3.3%
- pattern consistent with the 'retreat from hyper-globalisation' described in Extract B

0 3 Extract C identifies 'deadweight loss triangles in both production and consumption; loss of gains from specialisation'. With the help of a supply-and-demand diagram, analyse how the imposition of a tariff on imported goods reduces overall economic welfare. **[9 marks]**

Levels of response marking grid — 9-mark questions

The following generic grid should be used when marking all 9-mark questions in Section A (questions 0 3 and 0 7).

Level of response	Response	Max 9 marks
3	• is well organised and develops one or more of the key issues that are relevant to the question • shows sound knowledge and understanding of relevant economic terminology, concepts and principles • includes good application of relevant economic principles and/or good use of data to support the response • includes well-focused analysis with a clear, logical chain of reasoning • includes a relevant diagram that will, at the top of this level, be accurate and used appropriately.	7–9 marks
2	• includes one or more issues that are relevant to the question • shows reasonable knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles and/or data to the question • includes some reasonable analysis but it might not be adequately developed and may be confused in places • may include a relevant diagram.	4–6 marks
1	• is very brief and/or lacks coherence • shows some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • demonstrates very limited ability to apply relevant economic principles and/or data to the question • may include some very limited analysis but the analysis lacks focus and/or becomes confused • may include a relevant diagram but the diagram is not used and/or is inaccurate in some respects.	1–3 marks

Suggested diagram:

A standard tariff-welfare diagram is expected. Axes: price / quantity. Domestic supply (upward-sloping) and domestic demand (downward-sloping). Free-trade world price Pw lies below the domestic equilibrium price. At Pw, domestic producers supply Qs1; domestic consumers demand Qd1; imports = Qd1 – Qs1. Imposing a tariff t raises the domestic price to Pw + t. Domestic producers expand to Qs2 > Qs1; consumers reduce demand to Qd2 < Qd1; imports fall to Qd2 – Qs2. Consumer surplus falls by area A+B+C+D. Producer surplus rises by area A. Tariff revenue = area C. Deadweight losses are triangles B (production distortion — higher-cost domestic production displaces cheaper imports) and D (consumption distortion — consumers deterred by artificially high price). Total welfare loss = B + D.

Relevant issues include:

- definitions of tariff, consumer surplus, producer surplus, deadweight loss
- a tariff raises the domestic price above the world price by the amount of the tariff
- consumers lose: higher price and lower quantity = large reduction in consumer surplus
- producers gain: higher price allows expansion of domestic production — gain in producer surplus
- government gains: tariff revenue equals the tariff rate × imports under the tariff
- two deadweight loss triangles: production distortion (inefficient domestic production replacing cheaper imports) and consumption distortion (consumers deterred by artificial price)

- total welfare loss = consumer loss – producer gain – government revenue = sum of two DWL triangles
- empirical evidence from the 2018 US tariffs: welfare cost of around $800 per US household per year (Extract C)
- the result can be magnified in the long run through loss of dynamic efficiency and gains from specialisation
- retaliation by trading partners: foreign countries typically respond with their own tariffs, creating additional welfare losses on export side

0 4 Extract C states that 'in industries with large positive externalities... infant-industry protection may be justified.' Use the extracts and your knowledge of economics to evaluate the view that protectionist policies can ever be justified on economic grounds in advanced economies such as the UK and the United States. **[25 marks]**

Areas for discussion include:

- the standard case against protectionism: consumer welfare loss; deadweight loss; reduced gains from comparative advantage; loss of dynamic efficiency; risk of retaliation
- empirical evidence: 2018 US-China trade war produced large welfare losses, with incidence falling on US consumers ($800/household/year) rather than Chinese exporters as intended
- the case for strategic protectionism:
- (i) infant industry argument — temporary protection to allow young industries in advanced economies (e.g. semiconductors, green hydrogen, batteries) to achieve scale and compete
- (ii) positive externality argument — industries with large spillovers (R&D;, network effects) may be under-provided in free markets and justify subsidy or protection
- (iii) national security — protecting critical supply chains (semiconductors, energy, pharmaceuticals) from politically risky concentrations
- (iv) distributional argument — compensating workers and communities hurt by trade dislocation via redistribution is politically preferable to protectionism but may be insufficient in practice
- (v) second-best arguments — where other market failures exist (pollution abroad, labour standards), tariffs may partially offset
- the case against strategic protectionism: governments struggle to pick winners; industry lobbying captures policy; subsidies preferable to tariffs where positive externalities are the rationale; Article XX of GATT provides narrower exceptions than security tariffs
- modern industrial policy: US CHIPS Act ($280bn) and Inflation Reduction Act combine subsidies and local-content rules; EU Green Deal Industrial Plan; UK sovereign wealth / industrial strategy debates
- effects on the world trading system: unilateral measures risk retaliation, WTO disputes, fragmentation — IMF has warned of up to 7% of global output loss in a full trade war
- friend-shoring and near-shoring: arguably a rational risk-management response rather than protectionism in the traditional sense
- empirical evidence for / against: Japan and South Korea's development success involved strategic industrial policy; but this was in specific conditions (developing, catching-up economies) that advanced economies do not face
- government failure risks: regulatory capture, rent-seeking, persistence of protection beyond infant stage, corruption

Evaluation / judgement:

- a narrow economic case for protection exists (infant industry, externalities, national security) — but most of this is better addressed by subsidies than tariffs
- the broader political economy case (distributional consequences, community displacement) is real but again better addressed by redistribution than by trade restrictions
- protectionism in advanced economies typically imposes costs on consumers that exceed the gains to protected producers
- strongest answers distinguish carefully between tariffs and subsidies as policy tools, and reach a supported judgement recognising both the genuine market-failure arguments and the risks of government failure

Use the levels mark scheme for 25-mark questions to award candidates marks for this question.

Section A

Context 2 **Total for this Context: 40 marks**

Economic development and global inequality

0 5 Using the data in Extract D (Figure 3), calculate the ratio of the GNI per capita of the United Kingdom to that of Niger in 2023. Give your answer to one decimal place. **[2 marks]**

Calculation:

51,400 / 1,300 = **39.5** (i.e. UK GNI per capita is 39.5 times that of Niger)

Response	Max 2 marks
For the correct answer (39.5 (ratio of 39.5:1)) with correct working shown.	2 marks
For correct working with an arithmetic slip; OR correct numerical answer without working shown.	1 mark

0 6 Explain how the data in Extract D (Figure 4) suggest that economic growth rates have diverged between emerging Asia and sub-Saharan Africa since 2000. **[4 marks]**

Response	Max 4 marks
• includes evidence that clearly shows that growth rates have diverged between emerging Asia and sub-Saharan Africa since 2000. • clearly explains how this data is evidence that growth rates have diverged between emerging Asia and sub-Saharan Africa since 20	4 marks
• includes evidence that shows that growth rates have diverged between emerging Asia and sub-Saharan Africa since 2000. • unclear explanation of how this data is evidence that growth rates have diverged between emerging Asia and sub-Saharan Africa si	3 marks
• includes evidence that shows that growth rates have diverged between emerging Asia and sub-Saharan Africa since 2000. • limited or no explanation of how this data is evidence that growth rates have diverged between emerging Asia and sub-Saharan Afr	2 marks
• includes evidence that does not clearly show that growth rates have diverged between emerging Asia and sub-Saharan Africa since • weak or no explanation.	1 mark

Relevant issues include:

- emerging & developing Asia averaged 8.4% p.a. (2000–2010), 6.8% (2010–2019), and 4.8% (2020–2024) — well above the global average throughout
- sub-Saharan Africa averaged 5.9% p.a. (2000–2010) but slowed to 3.4% (2010–2019) and 3.1% (2020–2024)
- the gap in growth rates widened: Asia–Africa gap was 2.5pp in 2000–2010 but 3.4pp in 2010–2019 and 1.7pp in 2020–2024 (though Africa remained below Asia throughout)
- compounding effect: sustained higher growth in Asia means per-capita incomes have risen much faster, resulting in divergence in living standards
- population growth is faster in Africa, so per-capita growth divergence is even larger than headline rates suggest
- consistent with Extract E's observation that China alone has lifted over 800m out of extreme poverty, while SSA accounts for a rising share of global poverty

0 7 Extract E refers to the 'Lewis model [which] emphasises the reallocation of labour from low-productivity traditional sectors to higher-productivity modern sectors.' With the help of a diagram, analyse how investment in education and physical capital in a developing economy can increase long-run output and real GDP per capita. **[9 marks]**

Suggested diagram:

A PPF and/or AD/LRAS diagram is expected. PPF diagram: investment in education and physical capital shifts the PPF outward — more of both consumer goods and capital goods can be produced. AD/LRAS diagram: LRAS shifts rightward from LRAS1 to LRAS2; real GDP rises from Y1 to Y2 and price level falls. Candidates may also use a Solow-type diagram showing how higher capital-labour ratio raises output per worker. Lewis-model diagram (optional): subsistence wage in rural sector, drawing labour into modern (urban) sector with higher marginal productivity.

Relevant issues include:

- definition of long-run economic growth and real GDP per capita
- investment raises the capital stock (K/L ratio) — each worker has more capital to work with, raising labour productivity
- education raises human capital — higher skill levels raise the marginal product of labour
- both shift the LRAS rightward — higher potential output at any given price level
- PPF outward shift — the economy can produce more of all goods
- Lewis model: labour reallocates from traditional (low-productivity) to modern (higher-productivity) sectors as capital accumulation and skills grow
- Harrod-Domar: growth rate ≈ savings rate / capital-output ratio — higher savings / investment raises growth
- Solow-Swan: diminishing returns to capital mean long-run growth requires technological progress; human-capital extensions show why education is central
- real GDP per capita rises if output growth exceeds population growth — important in developing economies with high population growth
- evidence: East Asian successes (Japan, Korea, Taiwan, China) combined high saving, strong investment, education expansion — consistent with the models

0 8 Extract F states that 'without a substantial increase in development finance, capital investment, and debt relief, the Sustainable Development Goals for 2030 will be missed by a wide margin.' Use the extracts and your knowledge of economics to evaluate the most effective policies for accelerating economic development in low-income countries such as those in sub-Saharan Africa. **[25 marks]**

Areas for discussion include:

- the scale of the challenge: extreme poverty rates of 27–50% in poorer SSA economies (Ethiopia, Niger); HDI below 0.5; 60% of low-income countries in debt distress (Extract F)
- the case for more development finance: investment funds are below those needed for infrastructure, health and education; climate finance commitment of $100bn p.a. only partially met; multilateral institutions (World Bank, regional development banks) expanding lending
- the case for more debt relief: HIPC initiative of 1990s–2000s lifted major debt burdens; rising debt-to-GDP post-pandemic has restored the issue; G20 Common Framework is progressing slowly
- the case for trade-led development: East Asian model of export-led industrialisation; Aid for Trade programmes; preferential access to EU/US markets (Everything But Arms, AGOA)
- the case for institutional reform: strong property rights, rule of law, reduced corruption are prerequisites for sustained investment (Acemoglu/Robinson — 'Why Nations Fail')
- the case for human capital investment: education (particularly girls' education); health systems; nutrition — each has high returns in long-run growth
- role of microfinance and mobile money: Kenya's M-PESA, Bangladesh's Grameen Bank have successfully reached unbanked populations
- the Washington Consensus critique: trade liberalisation, privatisation and fiscal austerity in the 1990s delivered mixed results — structural adjustment sometimes failed where institutional preconditions were lacking
- developmental state model: South Korea, Taiwan and more recently Vietnam combined state guidance with market discipline
- resource curse vs resource blessing: commodity-rich SSA economies (Nigeria, Angola) have often underperformed; diversification is crucial
- climate finance and adaptation: 2022 Pakistan floods (4% of GDP lost) show climate vulnerability; "loss and damage" fund at COP28 is a partial response
- political economy constraints: governance, conflict, weak tax capacity limit what external finance can achieve
- controversy over aid: Easterly and Moyo have argued aid has been counterproductive; Sachs and others defend targeted aid

Evaluation / judgement:

- no single policy is 'most effective' — development requires a mix of investment, institutions, openness to trade, and macroeconomic stability
- the binding constraint varies by country: resource-cursed economies need diversification; conflict-affected states need security; extreme-poverty economies need basic health and education
- external support (finance, debt relief, market access) is necessary but not sufficient — domestic reform is essential
- strongest answers distinguish between constraints that dominate in different settings and reach a supported, country-sensitive judgement

Use the levels mark scheme for 25-mark questions to award candidates marks for this question.

Section B

Levels of response marking grid — 15-mark questions

The following generic grid should be used when marking all 15-mark questions in Section B (questions 0 9, 1 1 and 1 3).

Level of response	Response	Max 15 marks
3	A good response that: • is well organised and develops a selection of the key issues that are relevant to the question • shows sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response • includes well-focused analysis with clear, logical chains of reasoning.	11–15 marks
2	A reasonable response that: • focuses on issues that are relevant to the question • shows satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response • includes some reasonable analysis but which might not be adequately developed or becomes confused in places.	6–10 marks
1	A weak response that: • has identified one or more relevant issues • has some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • has very limited application of relevant economic principles to the given context and/or data to the question • might have some limited analysis but it may lack focus and/or become confused.	1–5 marks

Essay 1 — Total for this Essay: 40 marks

0 9 Explain, with the aid of an AD/AS diagram, how a combination of expansionary fiscal policy and contractionary monetary policy can affect the price level and real GDP in an economy. **[15 marks]**

Areas for discussion include:

- definitions: expansionary fiscal policy (higher G or lower T); contractionary monetary policy (higher interest rates, QT)
- the two policies pull AD in opposite directions: fiscal stimulus raises AD; monetary tightening reduces AD
- AD/AS diagram: initial equilibrium at PL1, Y1. If fiscal > monetary, AD shifts right, PL and Y rise; if monetary > fiscal, AD shifts left, PL and Y fall; if roughly balanced, small net effect on AD
- composition effects: fiscal stimulus typically boosts government/household spending; monetary tightening restrains investment and interest-sensitive consumption — net effect shifts the composition of demand
- the two policies can offset each other, a phenomenon often seen in 2022–24 in the UK, US and Eurozone as governments provided cost-of-living support while central banks raised rates
- impact on bond yields, exchange rate, and external balance: tight monetary + loose fiscal typically raises long-term yields and strengthens the currency (1980s US experience)
- effectiveness depends on relative size: in 2022–23 UK, monetary tightening was aggressive enough to produce near-recession despite some fiscal support
- distributional effects: fiscal policy can be targeted (transfers to low-income households) while monetary tightening tends to be regressive (hits mortgaged households hardest)

Use the levels mark scheme for 15-mark questions to award candidates marks for this question.

1 0 Evaluate the view that the coordination of fiscal and monetary policy is essential for maintaining macroeconomic stability in the UK. **[25 marks]**

Areas for discussion include:

- the case for coordination: avoids policy conflicts (one pulling up while the other pulls down); ensures efficient mix of instruments; 2022 'mini-budget' episode showed what happens when fiscal and monetary are at cross-purposes
- at the zero lower bound (as in 2009–2021), fiscal policy must do most of the work — coordination is essential for stabilisation
- coordinated response to major shocks: Covid-19 pandemic (2020) saw both fiscal support (furlough) and monetary easing (Bank Rate cut to 0.1%, QE expansion) working in the same direction
- the case against coordination: Bank of England independence since 1997 was explicitly designed to prevent political interference in monetary policy; coordination risks blurring this line
- Tinbergen principle: each instrument should target a clear objective; fiscal = output / distribution, monetary = inflation — separation improves accountability
- historical evidence: pre-1997 political interference in monetary policy was associated with higher inflation; post-1997 independence has anchored inflation expectations
- informal coordination works: the MPC's published forecasts condition the government's fiscal choices; fiscal announcements feed into the MPC's decisions
- formal coordination risks: if fiscal authorities effectively control monetary policy, inflation expectations become less anchored (fiscal dominance)
- examples of coordination failure: 2022 UK mini-budget produced a sharp rise in bond yields, forced MPC to tighten more than it otherwise would — costly for household incomes
- examples of good coordination: 2020 pandemic response; post-2008 QE + fiscal stimulus
- international comparison: US Fed and Treasury, ECB and Eurogroup all operate with formal independence but informal coordination
- the role of fiscal rules: 3% deficit ceiling, debt reduction by year 5 — help anchor expectations alongside inflation targeting

Evaluation / judgement:

- 'essential' is too strong — the UK has operated with independent monetary policy and separate fiscal accountability for 25+ years successfully
- but poor coordination can be very costly (2022 mini-budget)
- the optimal approach is independent institutions with transparent communication and credible frameworks — allowing informal coordination without sacrificing accountability
- strongest answers weigh the benefits of coordination against the risks of politicisation and reach a reasoned judgement

Use the levels mark scheme for 25-mark questions to award candidates marks for this question.

Essay 2 — Total for this Essay: 40 marks

1 1 Explain, with the aid of diagrams, how climate change represents both a market failure and a threat to long-run economic growth. **[15 marks]**

Areas for discussion include:

- definition of market failure: a misallocation of resources in a free market
- climate change as a negative externality of production — greenhouse gas emissions impose costs on third parties (global, long-term, intergenerational)
- MPC < MSC diagram: free market over-produces emission-intensive goods at Q1 > Q*; welfare loss triangle
- climate change also exhibits missing markets (the atmosphere has no price), information asymmetries, and collective-action failures — a 'super wicked' market failure
- threats to long-run growth: physical damage from extreme weather reduces productive capital; heat stress reduces labour productivity; water scarcity and crop failures harm agriculture; sea-level rise destroys infrastructure
- LRAS diagram: climate damage shifts LRAS leftward — lower potential output and higher price levels
- financial stability risks: stranded assets (fossil fuel reserves), asset-price re-pricing as carbon policies tighten, insurance-market disruption
- disproportionate impact on developing economies with limited adaptive capacity
- scale estimates: Stern Review (2006) estimated costs of inaction at 5–20% of global GDP; more recent estimates continue to highlight the tail risk of unchecked warming
- conversely, the green transition may offer investment and productivity gains — Net Zero technologies could shift LRAS rightward through innovation

Use the levels mark scheme for 15-mark questions to award candidates marks for this question.

1 2 Evaluate the view that policies to reduce greenhouse gas emissions will inevitably reduce the UK's rate of economic growth in the short run. **[25 marks]**

Areas for discussion include:

- the case for growth costs: carbon taxation and regulation raise costs for emission-intensive industries — energy, manufacturing, transport; pass-through raises consumer prices; some industries may contract; transition investment has opportunity cost
- empirical evidence: EU ETS has imposed modest costs on participating industries; UK carbon price support added to 2010s industrial energy costs
- the case against inevitable costs: green transition requires large investment in renewables, grid, electric vehicles, heat pumps — this IS economic activity, adding to AD and productive capacity
- Green New Deal arguments: green investment creates jobs (e.g. US IRA, EU Green Deal Industrial Plan)
- UK's specific advantages: offshore wind capacity, North Sea infrastructure, financial services (green finance), world-leading universities in relevant fields — transition could be a net opportunity
- distinction between gross and net effects: some sectors lose, others gain; net effect depends on policy design and investment response
- Porter hypothesis: well-designed environmental regulation can stimulate innovation and raise productivity
- short-run vs long-run: even if short-run growth is slower, the avoided climate damage preserves long-run growth capacity (LRAS argument)
- policy design matters critically: carbon pricing with revenue recycling is less growth-negative than regulation alone; targeted subsidies for clean technology can accelerate learning-by-doing
- avoiding carbon leakage: CBAM and similar measures prevent emissions simply moving abroad
- distributional and political issues: unevenly distributed costs (drivers, heating costs) create political resistance (2018 gilets jaunes in France; 2022 UK 'green crap' debate); redistribution is critical to sustain policy
- real-world evidence: UK has reduced emissions 50% since 1990 while GDP has grown 80% — decoupling is possible, though not guaranteed

• international dimension: unilateral UK action risks leakage and competitiveness loss without coordinated global action

Evaluation / judgement:

• "inevitably" is too strong — evidence shows decoupling is possible with good policy design

• short-run costs for some sectors are real but can be offset by investment in growing green sectors

• long-run, avoided climate damage more than offsets transition costs

• strongest answers distinguish sectors, time horizons, and policy designs, and reach a nuanced judgement

Use the levels mark scheme for 25-mark questions to award candidates marks for this question.

Essay 3 — Total for this Essay: 40 marks

1 3 Explain the main economic indicators used to measure the level of economic development in a country. **[15 marks]**

Areas for discussion include:

- definition: economic development = rising living standards, health, education, opportunity — broader than pure GDP growth
- GNI per capita (PPP-adjusted): monetary measure of average income; basis for World Bank country classifications
- limitations of GNI: ignores distribution, unpaid work, environmental degradation, quality of life
- Human Development Index (HDI): composite of GNI per capita, life expectancy, mean and expected years of schooling — addresses some GNI limitations
- Gini coefficient: measure of income inequality within countries
- poverty headcount: share of population below the extreme poverty line ($2.15/day, 2017 PPP) or national poverty line
- infant mortality, maternal mortality, life expectancy — health indicators correlated with development
- literacy rate and school enrolment — education indicators
- Multidimensional Poverty Index (MPI): deprivation across health, education, living standards
- Gender Inequality Index, ethnic/regional inequality measures
- environmental indicators: CO■ per capita, air quality, access to clean water and sanitation
- quality of governance and institutions: rule-of-law indices, corruption perceptions
- UK data application: Extract D shows how different indicators can give a coherent picture across countries

Use the levels mark scheme for 15-mark questions to award candidates marks for this question.

1 4 Evaluate the view that international trade is the most powerful force for reducing poverty in developing economies. **[25 marks]**

Areas for discussion include:

- the case for trade-led poverty reduction: East Asian export-led growth has lifted hundreds of millions out of extreme poverty (China, Vietnam, South Korea, Bangladesh)
- theoretical basis: comparative advantage, scale economies, technology transfer, FDI
- evidence: the countries with the fastest growth and poverty reduction have been those that have most increased their integration with world trade (McMillan/Rodrik)
- trade supports poverty reduction by: creating jobs in exporting industries; raising wages in tradable sectors; lowering prices for consumers via imports; attracting FDI and technology
- the case against trade as 'most powerful': trade benefits are uneven — regions and groups not connected to global markets (rural subsistence farmers) may not benefit; infant industries may be crushed by competition; volatility of commodity exports harms commodity-dependent poor countries
- alternative drivers of development: investment in health (improved life expectancy is a foundation for growth); education (human capital raises long-run productivity); institutions and governance (Acemoglu/Robinson — without rule of law, other policies fail); infrastructure; technology transfer (mobile phones, digital payments)
- sub-Saharan Africa's experience: many countries are highly trade-open (commodity exporters) yet have made slower poverty progress — suggesting trade alone is insufficient without complementary policies
- Washington Consensus critique: structural adjustment (including trade liberalisation) in the 1990s produced mixed results — institutional preconditions matter
- within-country inequality: trade liberalisation can raise within-country inequality even as it raises average incomes — poverty reduction depends on how gains are distributed
- role of aid, debt relief, climate finance: needed alongside trade opportunities
- geographic factors: landlocked, resource-poor economies face structural disadvantages in trade-led growth

- the policy framework for trade-led development: combine trade openness with investment in human capital, infrastructure, and institutions
- recent context: globalisation's slowdown (Extract A, B) may limit the trade-led route for current low-income economies; 'friend-shoring' and protectionism reduce opportunities

Evaluation / judgement:

- trade has been a powerful force for poverty reduction where conditions are right — but the conditions include institutions, human capital and infrastructure that are not delivered by trade alone
- "most powerful" is difficult to defend in all contexts — East Asia shows trade as central, while SSA shows trade is not sufficient without other reforms
- the binding constraint varies by country: trade opportunities matter more for economies with reasonable governance and infrastructure
- strongest answers distinguish between conditions and draw on regional evidence to reach a supported judgement

Use the levels mark scheme for 25-mark questions to award candidates marks for this question.

END OF MARK SCHEME